

Antioxidant-induced changes of the AP-1 transcription complex are paralleled by a selective suppression of human papillomavirus transcription.

Considering the involvement of a redox-regulatory pathway in the expression of human papillomaviruses (HPVs), HPV type 16 (HPV-16)-immortalized human keratinocytes were treated with the antioxidant pyrrolidine-dithiocarbamate (PDTC). PDTC induces elevated binding of the transcription factor AP-1 to its cognate recognition site within the viral regulatory region. Despite of increased AP-1 binding, normally indispensable for efficient HPV-16 transcription, viral gene expression was selectively suppressed at the level of initiation of transcription. Electrophoretic mobility supershift assays showed that the composition of the AP-1 complex, predominantly consisting of Jun homodimers in untreated cells, was altered. Irrespective of enhanced c-fos expression, c-jun was phosphorylated and became primarily heterodimerized with fra-1, which was also induced after PDTC incubation. Additionally, there was also an increased complex formation between c-jun and junB. Because both fra-1 and junB overexpression negatively interferes with c-jun/c-fos trans-activation of AP-1-responsive genes, our results suggest that the observed block in viral transcription is mainly the consequence of an antioxidant-induced reconstitution of the AP-1 transcription complex. Since expression of the c-jun/c-fos gene family is tightly regulated during cellular differentiation, defined reorganization of a central viral transcription factor may represent a novel mechanism controlling the transcription of pathogenic HPVs during keratinocyte differentiation and in the progression to cervical cancer.